WEEK -7

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 06-03-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 7)

B. **Sprint / Issue Goal:** System Architecture and Implementation logic.

C. **Status:** On Track

Backend logic and database interaction tasks are still under development

2. UML & OOAD Progress

A. **Diagrams Completed this week:**

1. Component Diagram: Visualized the physical modules of the system, including the Frontend (UI), Backend API (Flask), Machine Learning Model, and Database.
2. Deployment Diagram: Mapped the software components to hardware/nodes (e.g., cloud servers, user devices).

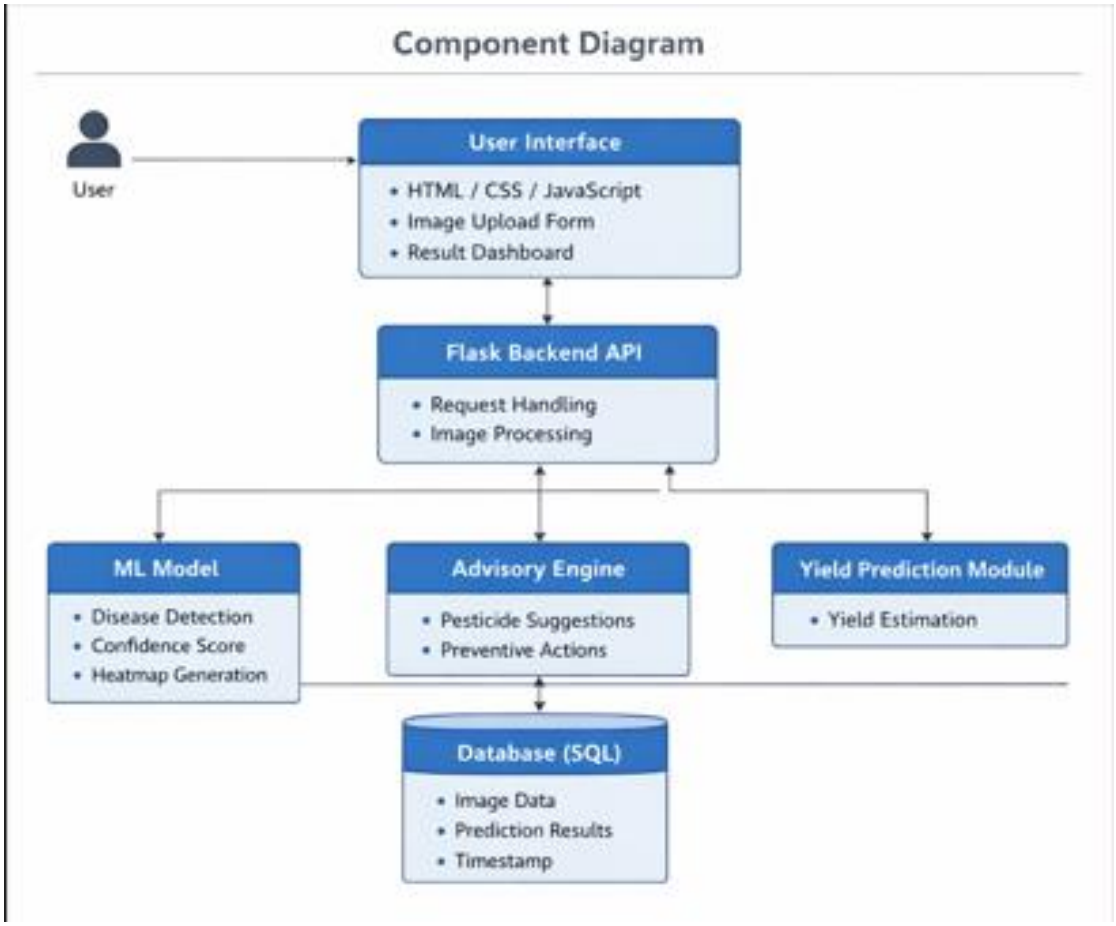
B. **Diagrams Revised:**

1. No revisions to diagrams were made this week.

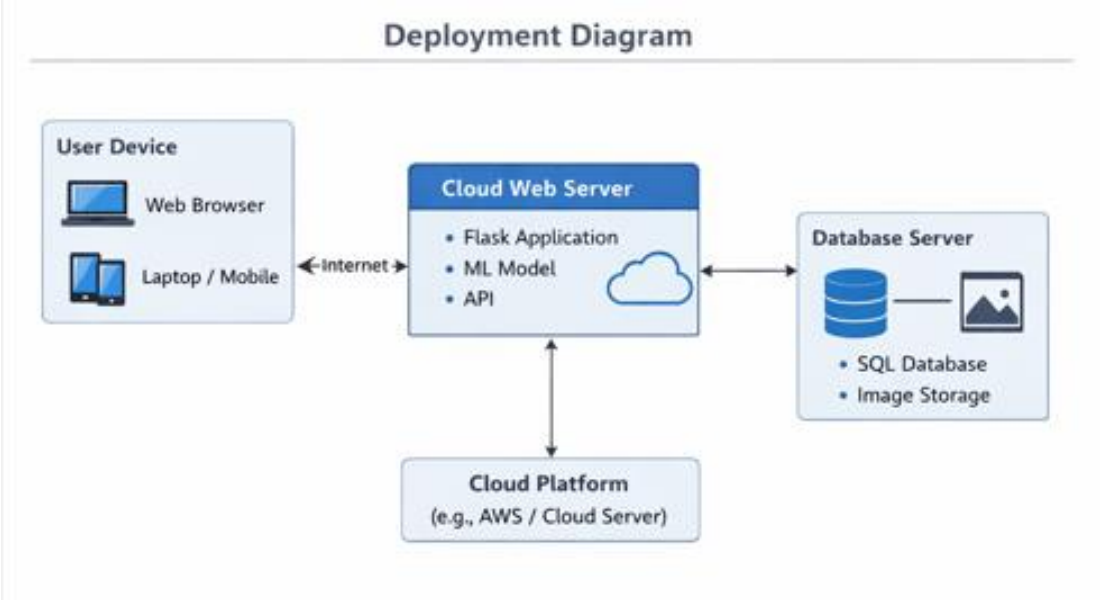
C. **OO Concept Applied:**

1. **Encapsulation**: Used in the Flask backend API for separating logic and database interaction.
2. **Modularization**: Components (like UI, backend API, database) were separated into distinct modules for easier management and implementation.

Component Diagram:



Deployment Diagram:



3.Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. **Task 1**: Frontend UI Development for image upload and display.
2. **Task 2**: Database Setup and CRUD operations for storing images and results.
3. **Task 3**: Component Diagram creation and validation.
4. **Task 4**: Deployment Diagram creation and validation.

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:**

1. CI/CD pipelines have been set up, focusing on:
   A. **Frontend**: Deployment and testing completed for the user interface.
   B. **Database**: Integrated with the frontend for image and result storage.
   C. **Backend**: No backend tasks or CI/CD configuration done yet. Work on the backend will begin in the following weeks.

4. Testing Summary

1. Total Unit Tests: Testing implemented for image uploading, frontend UI.
2. Pass Rate: Currently, 90% of the tests have passed, with only minor issues related to image upload size limitations.
3. Key Bug Identified: Minor bug in image upload handling: The system fails to process images above a certain size, which will be resolved by optimizing image size before upload.

5. Goals for Next Week

1. **Develop Core Logic Based on Class and Sequence Diagrams**:
   A. Design and implement core logic for image preprocessing, tree health monitoring, and fruit detection.
   B. Develop the logic based on the Class and Sequence Diagrams to define how components interact and how operations flow.
2. **Backend Logic Development**:
   A. Continue development of backend logic for image preprocessing, tree health monitoring, and fruit detection.
3. **Frontend Testing**:
   A. Further test the UI/UX with different image types and ensure smooth interaction between frontend, backend, and database.

Mentor Signature **Project Guide Signature**